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Key Points:

- Observations of rising frequency tones associated with the energy diffusion of proton shell distributions toward lower energies
- Linear theory applied to the observed distributions is qualitatively but not quantitatively consistent with the rising frequency
- Harmonic spacing of wave emissions suggests near source, but not conclusive if in source

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Event Study of Cyclic Diffusion of the Proton Shell Distribution Associated With Quasi-Periodic Rising Tone Fast Magnetosonic Waves

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Abstract It has been proposed that the cyclic rising tone in frequency observed in rising tone quasi-periodic (QP) fast magnetosonic waves is due to the cyclic diffusion of the inner energy edge of the source proton shell distribution toward lower energies. Linear theory shows that as the inner edge of the proton shell diffuses inwards the unstable frequency rises leading to a rising tone. We present the first observations of this diffusion of the proton shell distribution for this type of wave event, observed by the HOPE instrument on Van Allen Probes B, where this diffusion is clearly observed in two back-to-back QP wave cycles. Linear instability is performed on the observed proton shell distribution, and the linear growth rate is compared with the observed wave frequency; the linear theory applied to the observed proton shell distributions was qualitatively but not quantitatively consistent with the observed wave frequencies. Nearby Waveform Receiver burst mode measurements showed that the spacing of the emission bands in frequency was very close to the proton cyclotron frequency f_{cP} , suggesting a near source. However, a ± 2 nT spacecraft induced oscillation at $\frac{1}{2}$ the spacecraft spin period in the background magnetic field magnitude could not allow us to conclude that the emission bands were at integer multiples of f_{cP} , a stronger condition for source proximity.

Plain Language Summary Nature tends to head toward a steady state solution, but this state is rarely reached and instead nature fluctuates about the steady state. Often the fluctuations are quasi-periodic; for example, waves modulating clouds in the sky often have a quasi-periodic appearance. In plasmas this quasi-periodic phenomenon is sometimes observed in fast magnetosonic waves that are observed in the inner magnetosphere near the magnetic equator. These waves are one of the dominant wave modes in the Earth's inner magnetosphere and strongly interact with the lower proton energy part of the ring current and are believed to have a moderate role in scattering of radiation belt electrons. When these waves exhibit quasi-periodic behavior, the frequency rises during each quasi-period, and restarts at a lower frequency at the start of the next quasi-period. This event study is the first paper to attempt to observationally characterize the diffusion to lower energies of the proton source of these waves as the wave generation frequency rises during one quasi-period.

1. Introduction

Fast magnetosonic waves (typically $f > f_{cP}$ and $f < f_{LHR}$), where f is the wave frequency, f_{cP} is the proton cyclotron frequency and f_{LHR} is the lower hybrid resonance frequency) are one of the dominant wave modes in the Earth's inner magnetosphere and are chiefly observed around the magnetic equator (e.g., Boardsen et al., 2016; Meredith et al., 2008; Perraut et al., 1982). They play a moderate role in the energization and scattering of radiation belt electrons (Bortnik & Thorne, 2010; Li et al., 2014; Ni et al., 2017; Xiao et al., 2015) and strongly interact with the low energy edge of the proton ring current (Yuan et al., 2020). Proton shell (also termed ring) distributions have been shown to be the source of these waves where they are typically generated at frequencies at harmonics of the f_{cP} where the ion Bernstein mode couples to the whistler mode at wave normal angles near 90° (e.g., Chen et al., 2010; Gary et al., 2010; Sun et al., 2016). Observational analysis (Boardsen et al., 2018) and kinetic simulations in a dipole geometry (Min et al., 2018, 2020) for a case study have shown that during wave growth in the source region, wave propagation is primarily in the azimuthal direction as opposed to the radial direction. These waves can propagate large distances (multiple Earth radii) away from the source region with little

attenuation (e.g., Horne et al., 2000). Although not a focus of this study, we note that sub-proton cyclotron harmonic tones of the fast magnetosonic mode have recently been observed, whose spacing is much less than f_{cp} (Li et al., 2024; Wang et al., 2025). The study by Li et al., 2024 showed the existence of sub-harmonic emissions of rising tones whose frequency spacing was much less than f_{cp} . The study of Wang et al. (2025) for an event in which QP rising tones spaced at f_{cp} demonstrated that for that event the harmonics of f_{cp} are composed of sub-harmonics spaced at the O+ cyclotron frequency.

Roughly 5% (Němec et al., 2015, 2018, 2020) of these wave events exhibit quasi-period behavior and this behavior is sorted into two categories: non-dispersive and dispersive wave tones. For the non-dispersive category all frequencies are excited at once and the quasi-period of these tones are associated with and equal to the period of ULF waves. The non-dispersive category has been explained in terms of the ULF wave adiabatic modulation of the proton shell distribution turning on and off the instability (Zhu et al., 2019).

The dispersive category consists of quasi-periodic (QP) repeated rising tones, where each tone is composed of harmonics near $n f_{cp}$ (see Figure 8 of Němec et al., 2015; Figure 2a of Němec et al., 2020), which starts at the lower harmonics (say $n = 10$) and rises to higher harmonics (say $n = 36$) as time increases. This category is not associated with ULF waves and is believed to arise in a homogenous source region (Boardsen, Hospodarsky, et al., 2014; Němec et al., 2015). Non-linear (Fu et al., 2014) and quasi-linear (Boardsen, Hospodarsky, et al., 2014) mechanisms have been proposed as possible explanations. The latter proposed that this dispersive QP modulation is analogous to the dispersive QP behavior of electron whistler waves based on quasi-linear theory (Pasmanik et al., 2004a, 2004b; Trakhtengerts, 1995). In those studies, the electron whistler waves whose wave vector ($\mathbf{k}$) is parallel to the background magnetic field ($\mathbf{B}$) are generated by electron loss cone distributions; the generation of these waves leads to the pitch angle diffusion toward pitch angles of 0 and 180°, as the pitch angle changes linear theory predicts that the excited wave frequency increases with time. When the loss cone closes, the instability turns off and does not turn on until the loss cone is reformed. The quasi period is due to the interplay between the electron pitch angle diffusion and the reformation of the loss cone distribution in the source region. Boardsen, Hospodarsky, et al. (2014) proposed a similar mechanism where whistler waves with the $\mathbf{k}$ along $\mathbf{B}$, are replaced by fast magnetosonic waves that are azimuthally directed (direction of optimal gain) with $\mathbf{k}$ perpendicular to $\mathbf{B}$, and electron pitch angle diffusion of an electron loss cone distribution is replaced with energy diffusion of a proton shell distribution. Boardsen, Hospodarsky, et al. (2014) suggested that as the edge of the proton shell distribution diffuses toward lower energies, the linearly unstable frequency would increase, leading to a rising tone. Sun et al. (2020) showed, through kinetic simulation, starting with an unstable proton shell distribution, that as it evolves in time, the protons diffuse toward lower energy while the excited waves increase in frequency. From linear instability analysis of time slices of the proton shell throughout the simulation, Sun et al. (2020) showed that the unstable frequencies increase in time and are consistent with the excited frequencies in the simulation until the instability is quenched. We note that this was a relaxation simulation, so QP behavior, which may require replenishing protons of the unstable distribution via $\mathbf{E} \times \mathbf{B}$, gradient and curvature drifts, was not studied.

Diffusion of the proton shell toward lower energies with increasing frequency characteristic of the rising tone has not been shown to occur in observational data. In this study using Van Allen Probes-B we present a QP event where energy diffusion of the proton shell distribution is clearly observed with increasing wave frequency for two QP cycles, with fuzzy trends in neighboring cycles. We perform linear instability analysis using the observed proton shell VDF as input in both the resonant and non-resonant limits and compare it with the observed frequency of the emissions. The rest of the paper is structured as follows: Section 2 is on the data, Section 3 is on the observations, Section 4 applies linear instability analysis to the observations, followed by a discussion section and conclusion section.

2. Data

The event analyzed in this study uses Van Allen Probes-B data, over which we focus mainly on 2 QP cycles that are back-to-back. The QP emissions are detected using the EMFISIS Wave Form Receiver (WFR) 6 s cadence survey data, which is composed of 65 log spaced frequency channels from 2 Hz to 11 kHz (Kletzing, 2022a; Kletzing et al., 2013). The WFR survey data have sufficient frequency resolution to characterize the rising tone, but not enough to resolve the frequency bands near the harmonics of the proton cyclotron frequency that compose the rising tone. To resolve the harmonic structure, limited WFR burst mode data (Kletzing, 2022b) are available

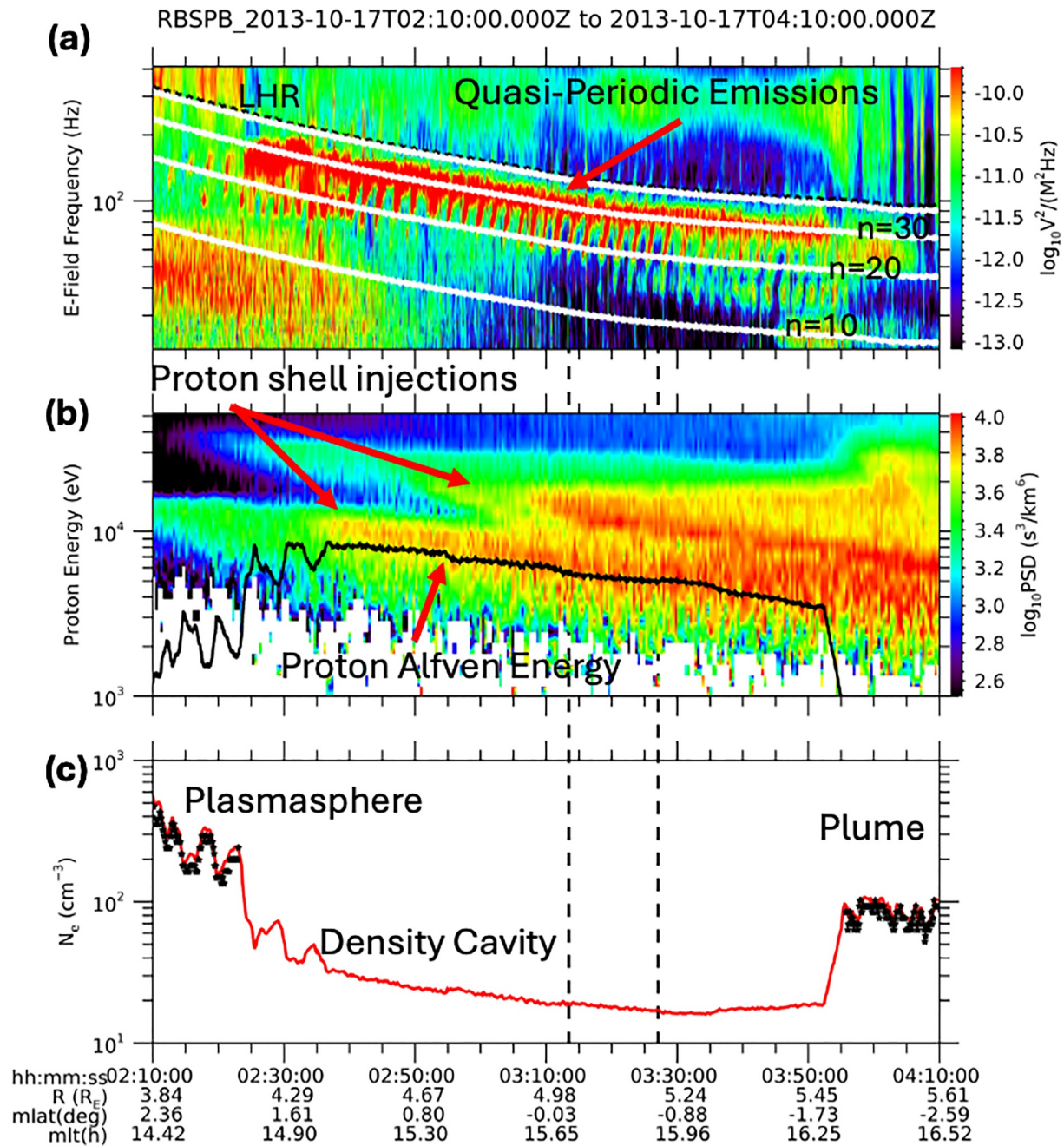


Figure 1. (a) Electric Field WFR spectrogram of dispersive quasi-periodic (QP) emissions with a QP of ~ 140 s. The white curves are at the 10th, 20th, 30th, and 40th harmonics of the proton cyclotron frequency. The black-dashed curve is at the Lower Hybrid Frequency (LHR). The fast magnetosonic mode QP emissions are primarily between the 10th harmonic and the 40th harmonics. Panel (b) 22 s cadence HOPE proton omnidirectional phase space density (PSD). Two proton injections are indicated, and a third occurs near the end of the interval. The black curve is the proton Alfvén energy. The proton Alfvén energy lies within the energy range of the lower energy proton shell distribution. Panel (c) Plasma density derived from EMFISIS UHR emissions (black asterisks) and EFW spacecraft potential (red curve) shows that the emissions are observed in a density cavity.

within 3 min (Nemec et al., 2020) of the 2 QP cycles that we focus on. Each 6 s burst is broken up into 3 2 s segments up which the fast Fourier transform is performed. After the FFT is performed, the data are further calibrated by applying amplitude and phase corrections (see Boardsen et al., 2018). The average power spectral matrix for E & B is separately formed from the sum of the 3 2 s segments. The choice of 2 s is somewhat arbitrary but was chosen to allow some spectral averaging to reduce the spectral noise and still have enough frequency

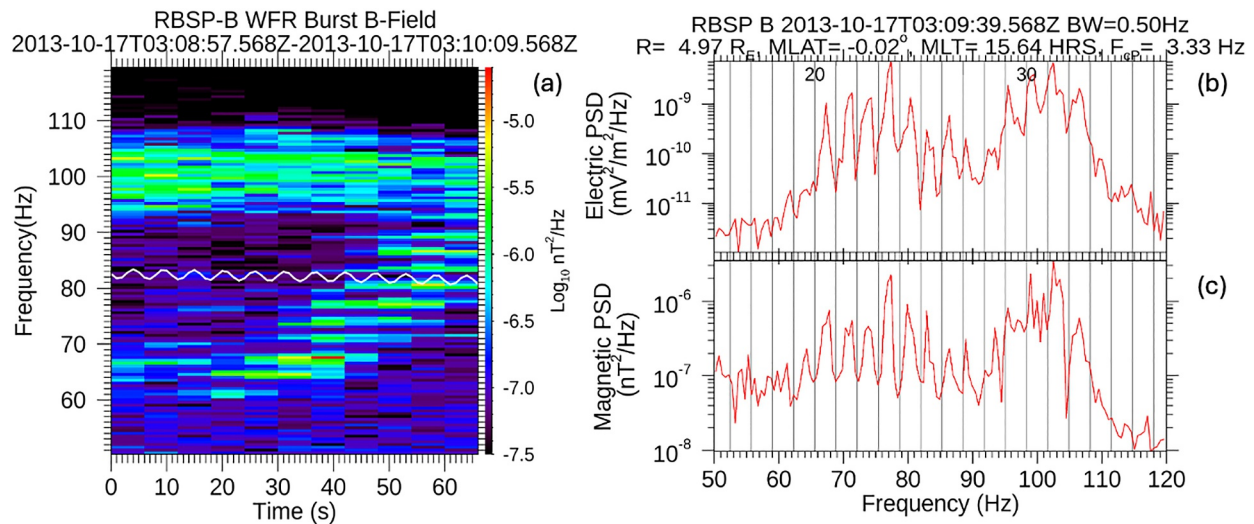


Figure 2. (a) WFR burst mode spectrogram of a rising tone (adapted from Nemec et al. (2020)). The oscillating white line is at the 25th f_{cp} harmonic, which indicates the range of uncertainty of the background magnetic field from which the harmonics are derived. Spectrum (a) of the burst mode electric field (b) and magnetic field (c). The vertical lines are at harmonics of the f_{cp} , for reference the 20th and 30th harmonics are labeled in (b). The f_{cp} values used to compute the harmonics in (c, b) are derived from the running average of the background magnetic field to remove the boom oscillations, see text for error estimates on location of the harmonics.

resolution of 0.5 Hz that is much less than f_{cp} . The trace of the power spectral matrix is shown in the data presented in this paper.

We note that besides EMFISIS, the Electric Fields and Waves (EFW) instrument has two burst mode products: “burst 1” for which calibrated level-2 data exist and are used in the studies by Li et al. (2024); Wang et al. (2025). No EFW “burst 1” data exists for the time interval of this study. For the time interval 2:10 to 3:09 UT, which is outside the analysis region, there are 25 EFW “burst 2” intervals of 4 s duration, but no calibrated level 2 data exists for this data set.

For proton phase space density measurements, referred to as the velocity distribution function (VDF), we use the HOPE instrument (Funsten, 2022; Funsten et al., 2013) proton data set, which has 72 log spaced energy channels from 1 eV to 51.8 keV, and 11 pitch angle bins for 4.5–175.5°, the cadence is 22 s and the time required to sample a complete distribution function is 11 s. HOPE sweeps up then down in energy with alternating energy bins, over this measurement interval.

For computing the proton cyclotron frequency (f_{cp}), 1 s Flux-Gate Magnetometer FGM data (Kletzing, 2022c) is linearly interpolated to the center times of the wave or proton measurements after performing a 2 spin period running average to removed spacecraft induced oscillations, as discussed in more detail in Section 3. For plasma density, the High Frequency Receiver (Kletzing et al., 2013) upper hybrid or continuum frequency cutoff derived plasma densities (Kurth et al., 2015, 2022) and the EFW instrument spacecraft potential derived plasma density (Wygant et al., 2022) are used, and the latter is calibrated using the HFR derived densities. To compute model derived average ion mass density, KP, AE, and DST indices from the OMNI (Papitashvili & King, 2020) data set are used.

3. Observations

Finding such QP events where diffusion could be observed is limited in number of events for at least three reasons. (a) The time required to make one proton phase space density measurement, and its cadence must be much less than the wave quasi-period, because one needs to study how the VDF changes with time as the emissions comprising the rising tone change with time. The quasi-period must be much greater than the HOPE cadence of 22 s. The quasi-period for the event of this paper is ~ 140 s, so 7 HOPE proton VDF measurements are made during each QP. (b) Sufficient energy channels must cover the proton shell distribution to resolve the diffusion. In the event the diffusion is observed from ~ 1 keV to about 11 keV covering 16 energy channels. (c)

The spacecraft must be in the source region; as noted, these emissions can propagate many Earth radii from the source. For the events analyzed in this study, we can only show that they are near source.

Figure 1 is an overview of the event. Figure 1a shows the WFR electric field spectrogram where QP emissions are observed between 2:20 and 3:50 UT on Oct 17th of 2013, they are observed between the 10th and the 30th harmonics of the local proton cyclotron frequency. The quasi-period is observed to be ~ 140 s for the repeating rising frequency tones. These waves are identified as fast magnetosonic by their high magnetic compressibility ~ 1 and wave normal angles near 90° (not shown) and occur chiefly between f_{cp} and lower hybrid frequency f_{LHR} . The f_{LHR} is indicated by the black dotted curve near the 40th harmonics. The 4 white curves are at the 10th, 20th, 30th, and 40th harmonics of the local proton cyclotron frequency.

For average ion atomic mass (M), we use the TrMne-3 trough model of Denton et al. (2025), which at 03:10:00 UT is $M = 1.44$ amu; we note that the rms error in M is large with a value of 1.40 amu. Denton et al. (2025) developed an equatorial electron density, mass density and average ion mass (M) models, dependent on location and KP, AE, and DST indices, for both the plasmasphere and plasmatrrough inferred from a data base (Denton, 2024) of the standing Alfvén wave frequencies and electron density from which the equatorial electron density and mass density were computed for each measurement from the Van Allen Probes Spacecraft and other missions. Before using the trough model, we verified that for Van Allen Probe B this database did not include data points for the time interval of this study, if it had those values would have been used for M instead of using the model cited. The plasma density with a mass composition of 94% protons, 4% He^+ and 2% O^+ is used in the computation of f_{LHR} . The 4% He^+ concentration was taken from the center of the scatter plot in Figure 2 of Craven et al. (1997) of the He^+/H^+ ratio at 4.67 RE. The O^+ concentration of 2% was used to match M . We note that there is an uncertainty in the relative concentrations, partly because the cold ion components cannot be observed by the Van Allen Probes instrumentation.

Note that there are weak repeating emissions extending above f_{LHR} ; this can occur for wave normal angles away from 90° (see Figure 1 of Boardsen et al. (2016)), this could be (a) a propagation effect, (b) source generation at f above f_{LHR} (Wu et al., 2022), or (c) due to wave-wave interaction and is not the topic of this paper. Figure 1b shows the HOPE omnidirectional phase space density (VDF) spectrogram; the proton shell distributions are indicated along with the proton Alfvén energy.

When the proton Alfvén energy lies on or near the proton shell distribution, this distribution can be unstable for the generation of this wave mode. The proton Alfvén energy lies within the energy ranges of the lower energy proton shell distribution. Note the strong decrease in the proton Alfvén energy inside the plasmasphere and plume, making this region unlikely candidates for wave generation for this event.

Figure 1c shows the density profile derived from the upper hybrid frequency (black asterisks) and the spacecraft potential (red curve). The plasmasphere, density cavity (trough), and plume are indicated, and fast magnetosonic QP emissions are observed in the cavity. We note that no EMFISIS derived densities are made in the cavity, so the spacecraft potential derived densities in the cavity are not adjusted to agree with EMFISIS derived density, in this regime the density is also a function of electron temperature so the relative measurement error in densities could be on the order of the $\sqrt{2}$ to 2. Using the Van Allen Probes spacecraft, Jahn et al. (2020) gave a factor of 2 when comparing spacecraft potential derived densities to upper hybrid resonance derived densities. Using POLAR spacecraft, Table 1 of Boardsen, Adrian, et al. (2014) which lists the spread of upper hybrid resonance derived densities against probe to spacecraft potential, the spread is about a factor of the $\sqrt{2}$ especially around a median density of 19 cm^{-3} . The density oscillations just inside the plasmasphere have a period of about 300 s, two rising tones are observed in this region and likely propagate in from the cavity, the spacing is also about 300 s, while in the density cavity the period is ~ 140 s, the cause of increase in period crossing the plasmopause is beyond the scope of this study. We believe this change is rare, for example, in Figures 2 and 4 of Boardsen, Hospodarsky, et al. (2014) no change in QP is observed while crossing the plasmopause; however, how frequently the QP changes when crossing the plasmopause should be explored in a follow-up study.

Figure 2a is a spectrogram (adapted from Nemec et al., 2020) of the WFR burst mode magnetic field data covering part of a rising tone. The white line is at the 25th harmonic of the proton cyclotron frequency, note the oscillation period is $\frac{1}{2}$ the spacecraft spin period and these oscillations are due to spacecraft induced oscillations of ± 2 nT, that are superimposed on the background magnetic field magnitude. For computing the cyclotron frequency used in Figure 2b, these oscillations are averaged out using a running average of 2 spacecraft spin periods. For the DC

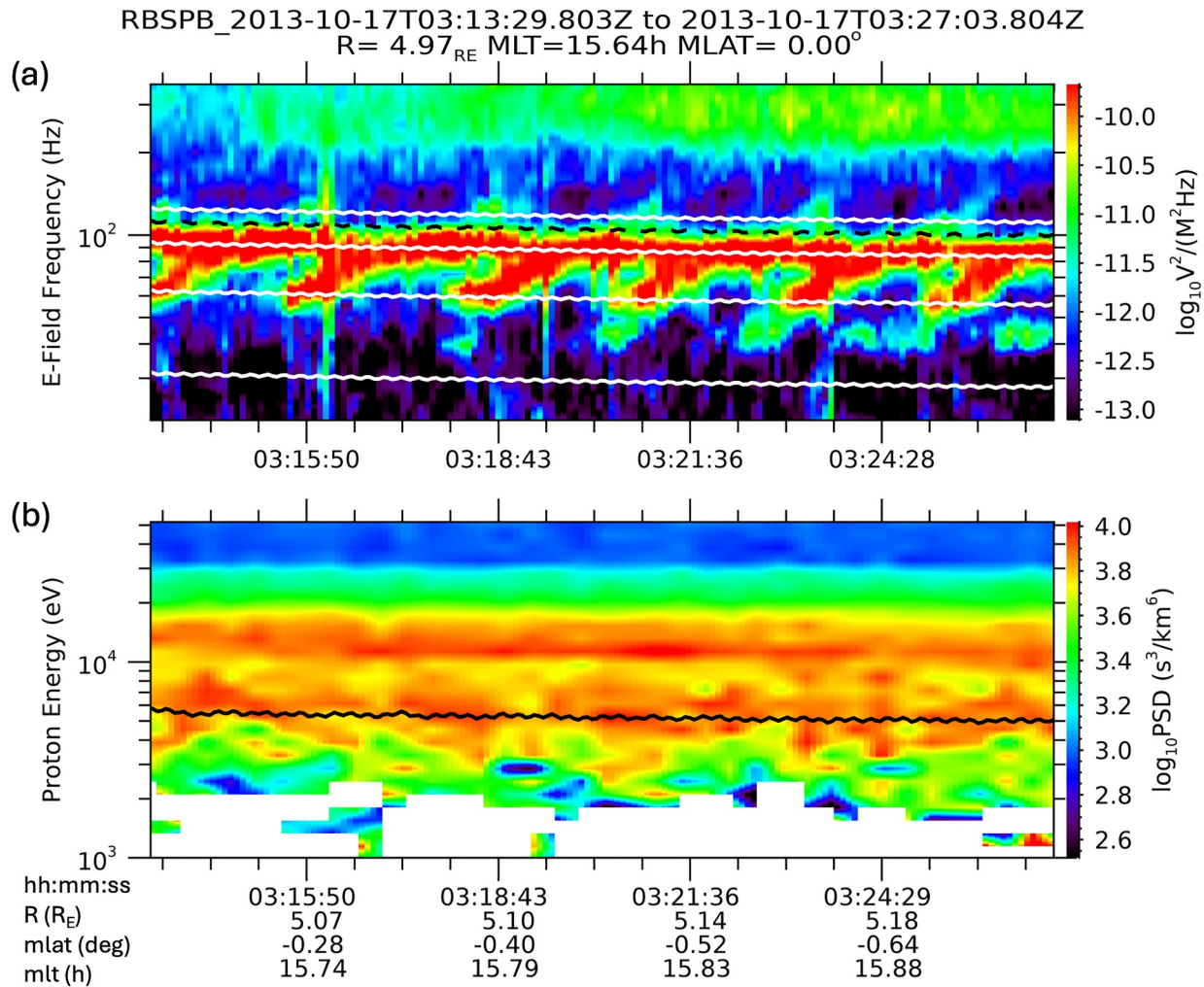


Figure 3. For the two temporally consecutive QP cycles shown on the left, (a) an increase in the frequency is accompanied by (b) a decrease in the lower-energy edge of the proton shell. In the following QP cycles shown on the right, this relationship becomes less organized. Note that the HOPE data shown here are interpolated.

magnetic field measurements, the digitization accuracy is 0.125 nT and the requirement is that the absolute error in the DC field should be 0.1 nT (Kletzing et al., 2013); therefore, we estimate the best scenario accuracy to be $0.1 + 0.125/2 = 0.16$ nT for the background magnetic field magnitude. Due to these oscillations, the accuracy of the background magnetic field is degraded and subsequently the accuracy of the value computed for f_{cp} . Figure 2b is a spectrum of the electric and magnetic field power spectral densities at the ~50 s mark of Figure 2a. The harmonics of f_{cp} using the average background magnetic field are shown, and the 20th and 30th harmonics are indicated in Figure 2b. For the best scenario accuracy of 0.16 nT, the error in the location (n db/B) of the 20th and 30th harmonics are ± 0.016 and $\pm 0.024 f_{cp}$ respectively, if the worst-case scenario of 2 nT is used, the error in the location of the harmonics are ± 0.2 and $\pm 0.3 f_{cp}$. There are two clusterings of emission bands in Figure 2a: the 20th to 24th harmonics and the 29th to 33rd harmonics; the former is from the previous rising tone. Even though the former emissions lie between the harmonics, the relative error for the worst-case scenario cannot exclude them having a sufficient overlap with the harmonics. Thus, at best we can conclude that the spacing between the harmonics is close to f_{cp} suggesting proximity to the source, but we cannot conclude that they satisfy a necessary condition to be in the source.

Zoom in of six quasi-periods is shown in Figure 3a of the electric field emissions and in Figure 3b of the HOPE proton shell distributions. The rising tone nature characteristic of these emissions is clearly observed in Figure 3a, while energy spread in the proton shell distribution appears to be modulated at the quasi-period and exhibits inward diffusion in energy, this is observed most clearly in the 2 QP cycles from ~03:14 to ~03:18 UT. A zoom in

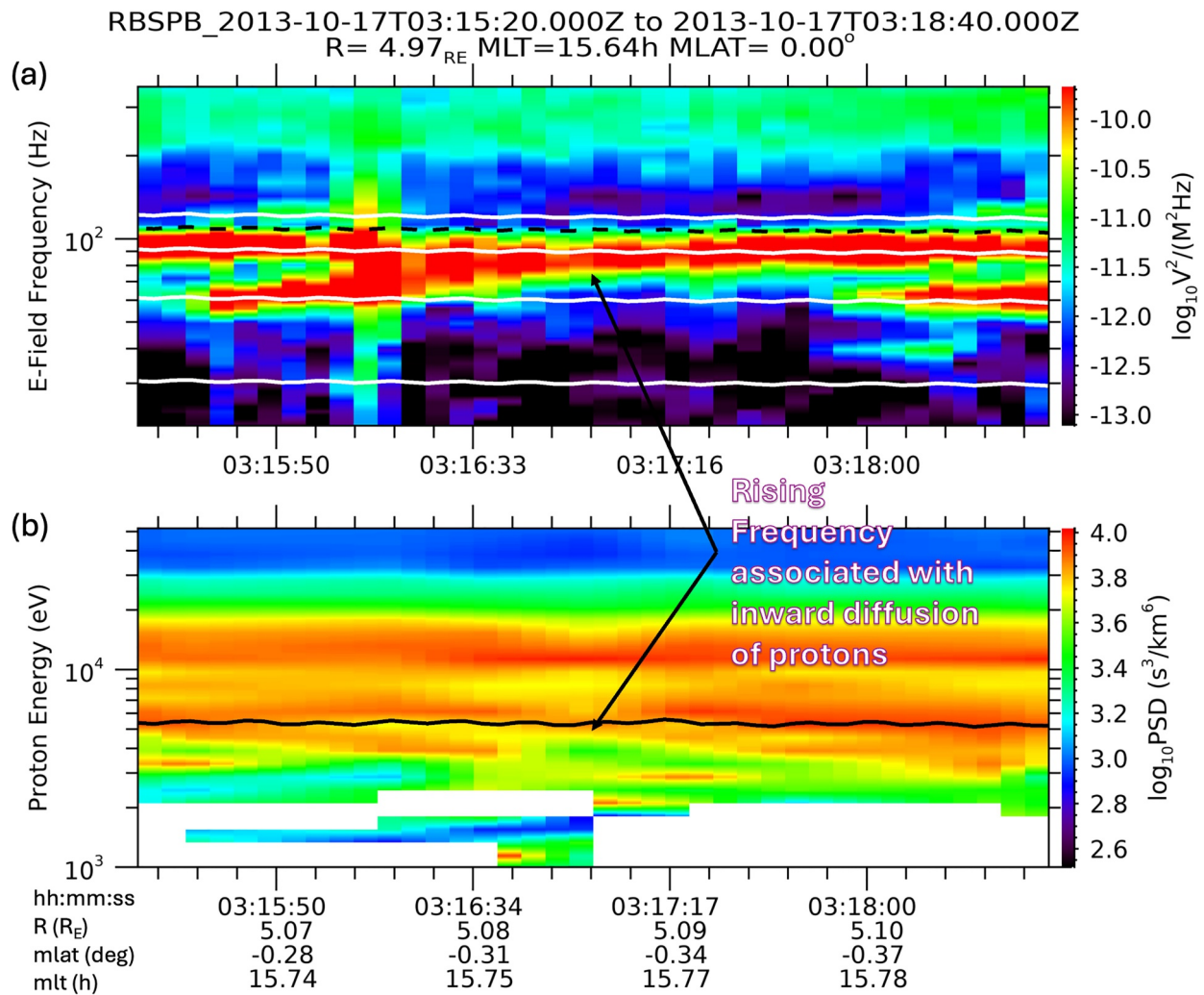


Figure 4. Zoom in of the QP interval in Figures 3a and 3b. The curves are defined in Figure 1. This interval clearly shows that as the emission frequency increases, the lower energy edge of the proton shell decreases and is the focus of the instability analysis. Note that HOPE data is interpolated.

of the cycle where this is most apparent is shown in Figure 4. The emission starts at low frequency around 03:15:15 UT and rises to higher frequency around 03:18:00 UT where the emission merges with those of the previous rising tones and another QP cycle begins. In Figure 3b, as the excited wave frequency rises, the lower energy edge of the proton shell distribution decreases. In the next section, linear instability analysis is applied to the observations to see how changes in the proton shell distribution can lead to the observed frequency change and whether qualitative or quantitative agreement can be achieved.

4. Linear Instability Analysis of Observations

In this section, we perform either resonant or non-resonant instability analysis on 2 proton HOPE VDF measurements for one rising tone. For resonant instability analysis, we use the formulas given in appendix A of Chen et al. (2010), in which corrections to the formulas given by Kennel (1966) have been made. The growth rate involves integration over a gyro-tropic distribution in $(V_{\parallel}, V_{\perp})$ space. For non-resonant instability analysis, we use the formulas given by Chen (2015), for which a reduced distribution is required, derived by integrating over $V_{\parallel}$. Both methods use cold plasma theory (Stix, 1992) for the real part of the dispersion relation. For a proton dominated plasma, in cold plasma theory the phase velocity is primarily a function of average mass density above the proton cyclotron frequency, and not very sensitive to the relative concentration of He⁺ and O⁺; therefore, the relative concentration has minimal impact on the linear growth rates.

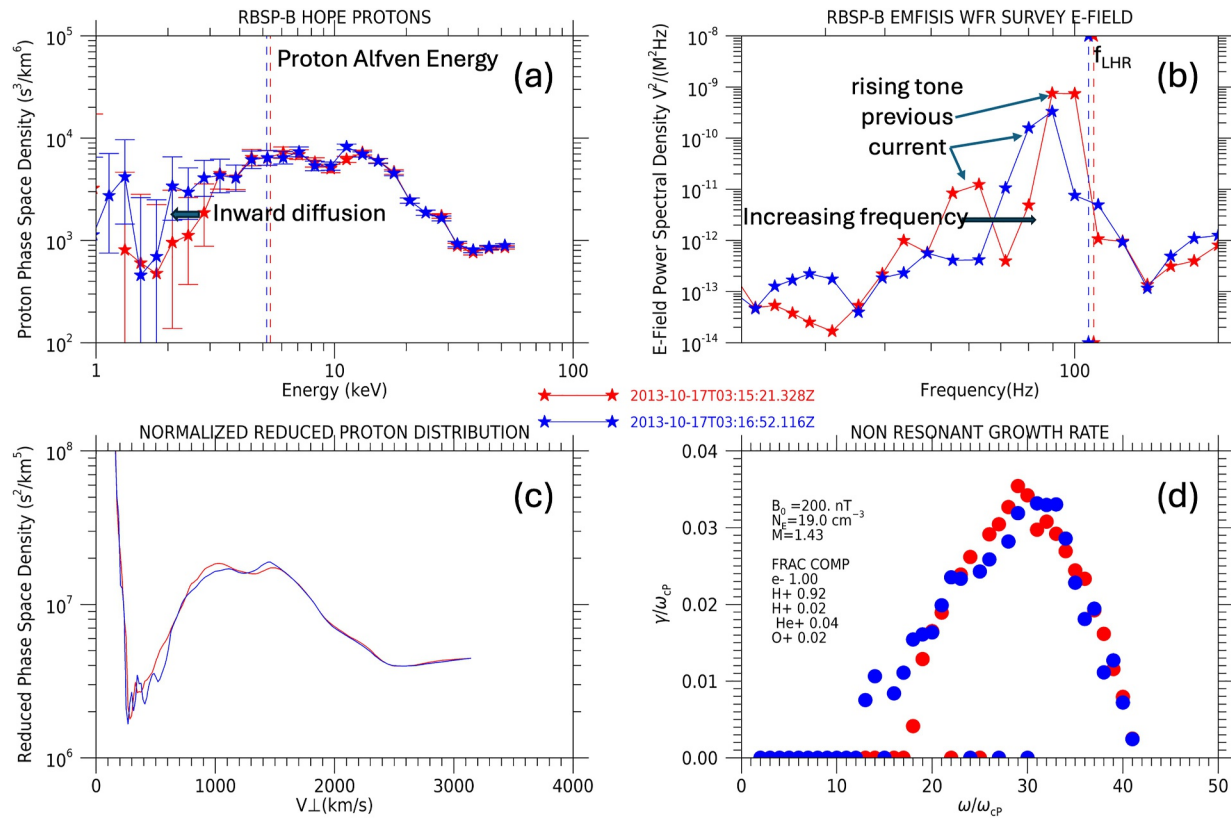


Figure 5. Comparison of the phase space density and WFR spectrum for two different times during a rising tone at 03:15:21 UT (red lines, dots, stars) which is early in the rising tone, and 03:16:52 UT (blue lines, dots, stars) which is later in the rising tone. Panel (a) proton phase space density versus energy spectra around a pitch angle of 90° , the vertical dashed lines are at the proton Alfvén energy at the two times, thick arrow indicates inward energy diffusion. Panel (b) 6 s WFR survey electric field spectra, the vertical dashed lines are at f_{LHR} at the two times, the arrows indicate emissions from the current and previous rising tone, thick arrow indicates frequency shift. Panel (c) is the reduced proton distribution function versus $V_\perp$, and (d) the non-resonant growth rate versus frequency normalized by the proton cyclotron frequency, computed from the reduced proton distribution; the parameters used are given in the legend in (d). For comparison of panels (b, d) the $f_{c,p}$ is about 3.39 Hz.

The HOPE proton VDF measurements in energy and pitch angle must be interpolated to a uniform grid in $(V_\parallel, V_\perp)$ for input into the resonant instability code and computing the reduced distribution function. We chose a 200×200 grid in velocity space with the range to be $\pm 3,100$ km/s for $V_\parallel$ and $0-3,100$ km/s for $V_\perp$. First, we spline interpolated the log of the HOPE VDFs in energy and pitch angle and then bilinearly interpolated the spline interpolated data to the 200×200 grid in $(V_\parallel, V_\perp)$. We interpolated the log of the VDF to minimize the surface curvature of the interpolated data; however, we did not attempt to fit the data within the Poisson error bars. The VDFs were normalized such that the integral over the VDF the distribution in SI units is 1, as required by Chen et al. (2010).

Can non-resonant instability analysis (Chen, 2015) of the reduced proton distribution function reproduce the observed frequencies of the rising tone? During one rising tone Figures 5 and 6 compare two times slices at 03:15:21 UT and 03:16:52 UT. In Figure 5, the red and blue lines, asterisks, and dots correspond to times of 03:15:21 UT and 03:16:52 UT. Figure 5a shows HOPE proton PSD measurements around pitch angles of 90° , the Poisson error bars are indicated, and the left-handed pointing thick arrow labeled “Inward diffusion” indicates what we interpret to be energy diffusion to lower energies. Figure 5b show two survey WFR spectra, the right-handed pointing thick arrow labeled ‘Increasing frequency’ indicates the frequency shift between these two times. The frequency of the bands rises from ~ 60 to ~ 85 Hz while the inner edge of the proton diffuses toward lower energies. The WFR frequencies of two spectral measurements composing the tone at 03:15:21 UT are 55.5 (the peak around 100 Hz is from the previous rising tone) and 63.0 Hz with proton harmonics numbers of 18.2 and 20.7 $f_{c,p}$, and at 03:16:52 UT are 80.1 and 89.7 Hz with proton harmonics numbers of 26.3 and 29.4 $f_{c,p}$. Figure 5(c) shows the reduced proton distribution function formed by integrating the proton phase space densities over the parallel velocity, unfortunately there is not clear signature of inward energy diffusion; between velocities of 600

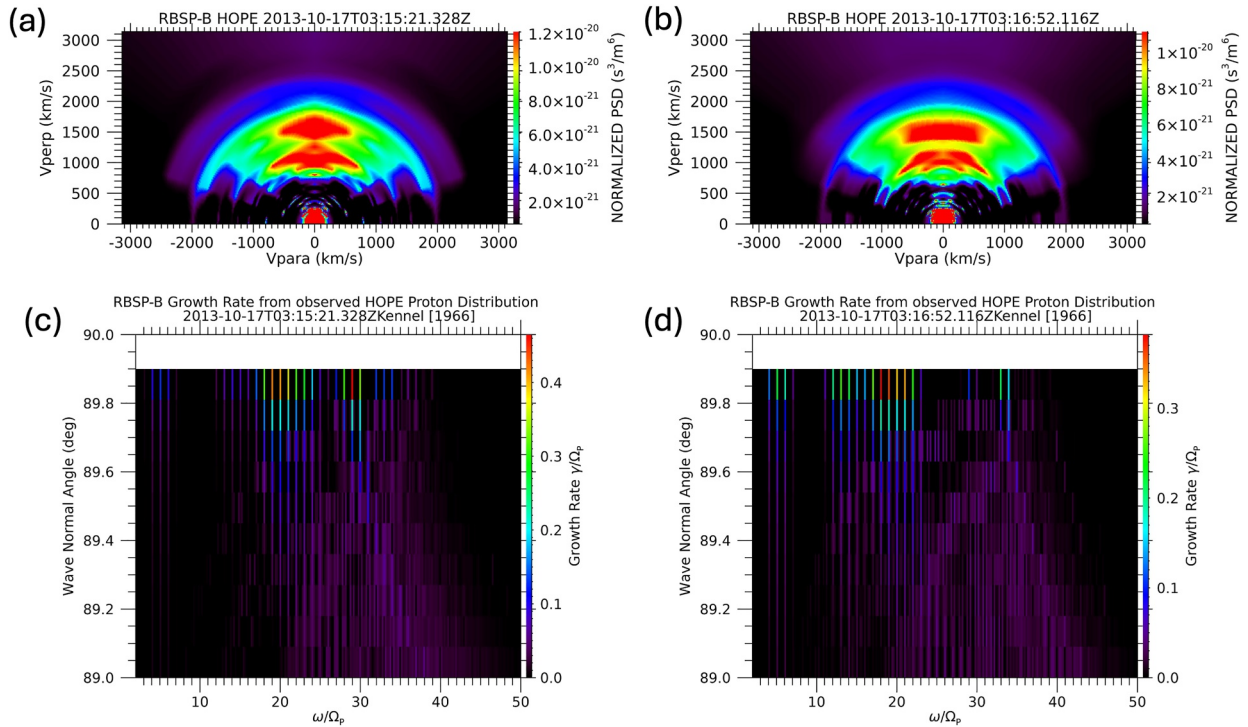


Figure 6. (a, b) Normalized ($V_{\parallel}$, $V_{\perp}$) Proton VDF slices, symmetrized about $V_{\parallel} = 0$, the two times at the top of each panel are the same as in Figure 5, the VDFs are normalized in SI units such that their integral over phase space is 1. Panels (c, d) Resonant growth rate spectrograms of the VDF of panels (a, b) (non-resonant effects not included) the similar background parameters in the legend of Figure 5d, the f_{cP} is about 3.39 Hz.

and 800 km/s, the reduced densities at the later time are at slightly lower velocities than those at the earlier time. The results of the non-resonant instability analysis are shown in Figure 5d, using the background magnetic field to compute f_{cP} , and the spacecraft potential derived densities of 19 cm^{-3} and model derived M of 1.43 amu . The peak non-resonant wave growth is $29 f_{cP}$ at 03:15:21 UT and 31 to $33 f_{cP}$ at 03:16:52 UT compared to the observed 19 and $28 f_{cP}$. The shifts in f_{cP} between observations and theory are $9 f_{cP}$ and $3 f_{cP}$, respectively, so theory does predict a rising frequency but is off by the number of f_{cP} .

The wave normal angle for the non-resonant analysis was 90° (Chen, 2015), with no resonant contribution ($k_{\parallel} = 0$). We performed resonant analysis (Chen et al., 2010) on the VDFs (Figures 6a and 6b) for the two times given in Figure 5. We perform resonant analysis only for wave normal angles between 89.5 and 89.95° using the observed HOPE proton distribution at the times in Figure 5, using background parameters given in Figure 5. The resonant growth rates are shown in Figures 6c and 6d for the VDFs in Figures 6a and 6b, respectively. Note that the resonant only growth rates will be larger than the full solution, which includes non-resonant effects. This is clearly shown in Figure 2 of Chen (2015), where the resonant only growth rates are larger than the growth rate computed using the full solution. In Chen (2015) exclusion of the non-resonance does not change the location in frequency of the peaks but changes the magnitude. For the bi-Maxwellian constructed shell distribution used in Chen (2015), the inclusion of non-resonant growth becomes important starting around 89.5° (see Figure 2 of Chen (2015)). To attempt to overcome this, we tried the code Leopard by Astfalk and Jenko (2017), which solves the full dispersion relation for arbitrary gyro-tropic VDFs in a warm non-relativistic plasma but could not get the code to converge for wave normal angles greater than 87° .

Comparing Figure 6c with Figure 6d positive-growth rate around $\sim 20 f_{cP}$ is observed; we believe this is due to the high energy proton shell distribution. We believe this does not show up in the non-resonant growth rates because the integral over $V_{\parallel}$ strongly reduces the positive slope of the peaks around 1,500 km/s observed in Figures 5b and 5e. Figures 6c and 6d shows positive-growth rates around harmonics 30 and $35 f_{cP}$, respectively, which are similar to the non-resonant computation.

5. Discussion

Here we discuss several factors that could contribute to the discrepancy between observation and theory (a) Use of the central measurement time of the proton measurements as opposed to the measurement times of the wave spectra could skew results, we estimate that at most this will be a 6 s error, estimating the rise time from the numbers in the previous paragraph $df_{cp}/dt \sim 0.1 f_{cp}/s$, for a 6 s time offset $\Delta f_{cp} \sim 0.6$, the shift is small but can be a contributing factor. (b) We note that both resonant and non-resonant growth rate peaks are highly dependent upon the Alfvén speed, which is a function of the magnetic field strength and the average plasma mass density, as noted the error bar in the model mass density is large and the spacecraft potential error can be on the order of the $\sqrt{2}$ to 2. If an electron plasma density of 24 cm^{-3} (higher mass density) is used, the peak non-resonant wave growth is $27 f_{cp}$ at 03:15:21 UT and 26 to $29 f_{cp}$ at 03:16:52 UT compared to the observed 19 and $28 f_{cp}$. Thus, agreement with theory is much better for the later, and only marginally better for the former. There is continuum radiation present during this interval, from which one could estimate the electron density from the continuum cutoff (e.g., Kurth et al., 2015), however the presence of strong electron cyclotron harmonics during this interval introduces ambiguity in identifying the continuum cutoff. (c) Lack of frequency resolution in the survey data is another contributing factor. (d) The instability analysis is not performed on an instantaneous distribution, but a distribution accumulated over 11 s over which the distribution is evolving. (e) An infinite number of smooth surfaces can be created that lie within the Poisson error bars of the discreet VDF measurements. The approach used in this study is likely not optimal.

For this event, the magnetic local time is near dusk, and its interpretation is complicated by the presence of two proton shell distributions. Even though studies have shown that these waves are generated outside the plasma-pause over magnetic local times from early morning through midnight to prenoon, the occurrence is strongly peaked around noon for both non-QP emissions and QP emissions; a good near noon time example with burst mode data and preferably one proton shell distribution is desirable. Potentially better candidates could be found than those shown in the study. The current study sets up some of the criteria for such candidate events.

6. Conclusion

In conclusion, a diffusive QP fast magnetosonic wave event is shown that we interpret to be associated with QP diffusion of the proton shell distribution. Two back-to-back wave cycles and the associated proton shell distribution clearly show that as the excited waves rise in frequency the inner edge of the proton shell distribution moves toward lower energy. This relationship is fuzzier in other neighboring QP cycles. However, we note using resonant and non-resonant theory that the computed growth rates whose frequencies increased during the rising tone did not match the observed wave frequency rise, but qualitatively the frequency variations are consistent. Hopefully, better events showing this effect will be discovered or the reanalysis of this event will improve the agreement.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The data on which this article is based are publicly available in Funsten (2022) for RBSP-B HOPE proton data (<https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspb/l3/ect/hope/pitchangle/rel04/>), Kletzing (2022a) for RBSP-B EMFISIS-WFR plasma wave spectral matrix data (<https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspb/l2/emfisis/wfr/spectral-matrix-diagonal-merged/>), Kletzing (2022b) for RBSP-B EMFISIS-WFR waveform data (<https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspb/l2/emfisis/wfr/waveform-continuous-burst/>), Kletzing (2022c) for RBSP-B fluxgate magnetometer data (<https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspb/l3/emfisis/magnetometer/1sec/sm/>), Kurth et al. (2022), for RBSP-B UHR derived electron density (<https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspb/l4/emfisis/density/>), Papitashvili and King (2020) for OMNI geomagnetic indices data (https://spdf.gsfc.nasa.gov/pub/data/omni/omni_cdaweb/hourly/), and Wygant et al. (2022) for RBSP-B EFW spacecraft potential derived electron density data (<https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspb/l3/efw/>). We thank the Space Physics Data Facility, which posts the data on the CDAWeb, for providing the data.

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